

Bimanual VLA Dinner-Table

Dual SO-101 arms set a dinner table in MuJoCo from a spoken command.

Runs on Intel CPU+iGPU via OpenVINO. Simulation-first, fully reproducible.

Problem & Solution

Table-setting needs two arms, language understanding, and edge deployment — rarely seen together.

We combine a language planner, a learned policy, bimanual coordination, and OpenVINO on in one repo.

How It Works

Planner splits the command into 6 phases; arms execute with atomic handoff ownership clamps.

Stack: MuJoCo, PyTorch BC policy (OpenVINO IR FP32/FP16), headless cameras, JSO

Demo: What You See

Screen recording: natural-language command, 10 randomized seeds, coordinated hands

HUD shows seed, outcome, phase, and the fill-meter proxy disclaimer on every clip.

Measured Results

Oracle baseline 8/10 full tasks (all non-pour phases 10/10); learned policy honestly 0/10 al.

Inference: CPU 0.08ms, iGPU 0.44ms FP16; torch parity MSE $6.8e-9$. All figures from our n.

Market Scope

TAM: home-assistive and hospitality robotics, a multi-billion-dollar emerging market.

SAM: simulation-first manipulation R&D teams needing cheap bimanual benchmarks on hardware.

Revenue & Competition

Paths: edge-deployment tooling licenses, simulation content packs, integration services.
Vs single-arm demos and cloud-GPU policies: ours is bimanual, language-driven, and ruggedity Intel chips.

Future Prospects

Scale to full cutlery sets, DAgger/ACT and SmolVLA finetuning, NPU targets on Core Ultra
Same pipeline ports to other tabletop tasks with new data only.

Reproducibility

One repo: build script regenerates the scene, one command per phase, JSON-verified commands.
Judges rerun sim, bench, and eval locally; no hidden steps, no projected numbers.

Thank You

Code, video, benchmark, and architecture summary are in the submission.

Honest partials today; the pipeline is built for full autonomy next.